

SPRINT 2

Stroke Prediction Analytics

Exploratory Data Analysis & Visualization

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Batch: 544-A

Dataset: cleaned healthcare_dataset.csv

1. Sprint 2 Overview

Sprint 2 focuses on Exploratory Data Analysis (EDA) and Visualization using the cleaned healthcare stroke dataset. The notebook's stated goal is to discover meaningful patterns, trends, and relationships through statistical analysis, Pandas operations, visualization, and business storytelling.

The clean dataset contains 5,110 patient records and 12 variables. The target variable is stroke: 0 = No stroke and 1 = Stroke. There are 249 stroke cases (4.87%) and 4,861 non-stroke cases (95.13%).

2. Business Understanding

Business problem: identify patient characteristics and health-related factors associated with stroke occurrence.

Business Question	Sprint 2 Analysis
How imbalanced is the stroke target?	Target distribution and percentage analysis
Which patient characteristics differ between stroke and non-stroke groups?	Group statistics, box plots, violin plots and pivot tables
How does stroke rate vary by age group?	Age-group aggregation
Which smoking categories have higher observed stroke rates?	Smoking-status group analysis
How do hypertension and heart disease jointly relate to stroke?	Joint crosstab and percentage analysis

Important interpretation rule: EDA findings describe observed associations in this dataset. They should not be treated as medical diagnoses or causal conclusions.

3. Import Libraries and Dataset

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_theme(style="whitegrid")
pd.set_option("display.max_columns", None)

df = pd.read_csv("/content/cleaned_healthcare_dataset")
```

NumPy is used for numerical operations such as infinity values and array operations. Pandas is used for loading and analyzing the DataFrame. Matplotlib and Seaborn are used for charts. The notebook sets a white-grid visualization theme and allows all columns to be displayed.

```
print("Shape:", df.shape)
display(df.head())
display(df.info())
```

shape confirms the dataset dimensions. head() shows the first five observations. info() provides column names, data types and non-null counts.

Metric	Clean Dataset Result
Rows	5,110
Columns	12
Missing cells	0
Duplicate rows	0
Target	stroke

4. Data Structure and Quality Check

```
print(df.columns.tolist())
print(df.dtypes)
print(df.isnull().sum())
print("Duplicate rows:", df.duplicated().sum())
```

Column	Data Type	Role
id	int64	Patient identifier
gender	object	Categorical
age	float64	Numerical
hypertension	int64	Binary health feature
heart_disease	int64	Binary health feature
ever_married	object	Categorical
work_type	object	Categorical
Residence_type	object	Categorical
avg_glucose_level	float64	Numerical
bmi	float64	Numerical
smoking_status	object	Categorical
stroke	int64	Target

The cleaned dataset has zero missing values and zero duplicate rows. Therefore, Sprint 2 analysis can proceed without repeating Sprint 1 missing-value treatment.

5. Descriptive Analysis – Numerical Summary

```
numerical_features = [
    "age", "hypertension", "heart_disease",
    "avg_glucose_level", "bmi", "stroke"
]

descriptive_stats = df[numerical_features].describe().T
descriptive_stats["median"] = df[numerical_features].median()
descriptive_stats["variance"] = df[numerical_features].var()
descriptive_stats["range"] = (
    df[numerical_features].max() - df[numerical_features].min()
)
descriptive_stats["skewness"] = df[numerical_features].skew()

display(descriptive_stats)
```

describe() supplies count, mean, standard deviation, minimum, quartiles and maximum. The notebook then adds median, variance, range and skewness. This creates a richer numerical profile than describe() alone.

Variable	Count	Mean	Median	Std Dev	Variance	Range	Skewness
age	5110	43.227	45.000	22.613	511.332	81.920	-0.137
hypertension	5110	0.097	0.000	0.297	0.088	1.000	2.715
heart_disease	5110	0.054	0.000	0.226	0.051	1.000	3.947
avg_glucose_level	5110	106.148	91.885	45.284	2050.601	216.620	1.572
bmi	5110	28.893	28.400	7.698	59.259	87.300	1.077
stroke	5110	0.049	0.000	0.215	0.046	1.000	4.193

Interpretation of Key Metrics

Mean and median represent central tendency. Standard deviation and variance describe dispersion. Range is maximum minus minimum. Skewness describes asymmetry. Positive skewness indicates a longer right tail.

Variable	Metric Observation
Age	Mean $\approx$ 43.23, median = 45.00; skewness $\approx$ -0.14
Avg glucose	Mean $\approx$ 106.15, median $\approx$ 91.89; positive skewness $\approx$ 1.57
BMI	Mean $\approx$ 28.89, median = 28.40; positive skewness $\approx$ 1.08
Hypertension	Mean $\approx$ 0.097; because it is 0/1, this corresponds to about 9.75% coded 1
Heart disease	Mean $\approx$ 0.054; about 5.40% coded 1
Stroke	Mean $\approx$ 0.0487; about 4.87% coded 1

6. Central Tendency and Dispersion

```
central_tendency = pd.DataFrame({
    "Mean": df[numerical_features].mean(),
    "Median": df[numerical_features].median(),
    "Mode": df[numerical_features].mode().iloc[0]
})

dispersion = pd.DataFrame({
    "Std Dev": df[numerical_features].std(),
    "Variance": df[numerical_features].var(),
    "Range": df[numerical_features].max() -
        df[numerical_features].min()
})

display(central_tendency)
display(dispersion)
```

The code creates two separate tables. `central_tendency` contains mean, median and mode. `dispersion` contains standard deviation, variance and range. Separating these concepts makes the statistical story easier to communicate.

7. Univariate Analysis

7.1 Numerical Distributions

```
continuous = ["age", "avg_glucose_level", "bmi"]

for col in continuous:
    fig, ax = plt.subplots(figsize=(7,4))
    sns.histplot(df[col], kde=True, ax=ax)
    ax.set_title(f"Distribution of {col}")
    ax.set_xlabel(col)
```

```
ax.set_ylabel("Frequency")
plt.show()
```

A histogram shows how observations are distributed across numerical values. KDE adds a smoothed estimate of the distribution. The notebook uses this for age, average glucose level and BMI.

7.2 Numerical Outlier View

```
for col in continuous:
    fig, ax = plt.subplots(figsize=(7,3.5))
    sns.boxplot(x=df[col], ax=ax)
    ax.set_title(f"Box Plot of {col}")
    plt.show()
```

A box plot summarizes median, quartiles and potential extreme observations. In Sprint 2 it is used as a visual complement to the numerical distribution analysis.

7.3 Categorical Frequencies

```
categorical_features = [
    "gender", "ever_married", "work_type",
    "Residence_type", "smoking_status",
    "hypertension", "heart_disease", "stroke"
]

for col in categorical_features:
    print(f"--- {col} ---")
    display(df[col].value_counts(dropna=False).to_frame("count"))
    display(
        (df[col].value_counts(normalize=True, dropna=False) * 100)
        .round(2).to_frame("percentage")
    )
```

`value_counts()` provides category counts. `normalize=True` converts counts to proportions, which are then multiplied by 100 to produce percentages. This allows categories with different sample sizes to be compared more fairly.

8. Target Distribution – Stroke

```
stroke_count = df["stroke"].value_counts()
stroke_percent = df["stroke"].value_counts(normalize=True) * 100
print(stroke_count)
print(stroke_percent)
```

Target	Count	Percentage
0 – No Stroke	4861	95.13%
1 – Stroke	249	4.87%

The dataset is highly imbalanced. Only 249 of 5,110 patients are stroke cases. This imbalance is important for later predictive modeling because a model could achieve high accuracy while performing poorly on the minority stroke class.

9. Bivariate Analysis

9.1 Numerical vs Numerical – Correlation and Scatter Plot

```
display(df[continuous].corr())

fig, ax = plt.subplots(figsize=(7,5))
```

```
sns.scatterplot(
    data=df,
    x="age",
    y="avg_glucose_level",
    hue="stroke",
    alpha=0.5,
    ax=ax
)
ax.set_title("Age vs Average Glucose Level by Stroke")
plt.show()
```

The correlation matrix measures linear association between continuous variables. The scatter plot adds visual context and colors observations by stroke outcome.

Relationship	Correlation
Age vs Avg glucose	0.238
Age vs BMI	0.326
Avg glucose vs BMI	0.169

The correlations among the continuous variables are positive but not extremely strong. This means the variables contain related information, but they do not move together perfectly.

9.2 Age and Stroke

```
df.groupby("stroke")["age"].agg(
    ["count", "mean", "median", "std", "min", "max"]
)
```

Stroke	Count	Mean Age	Median	Std Dev	Min	Max
0	4861	41.97	43.00	22.29	0.08	82.00
1	249	67.73	71.00	12.73	1.32	82.00

Observed metric: the stroke group has a much higher mean age (≈ 67.73) than the non-stroke group (≈ 41.97). This is an observed association in the dataset and is not by itself evidence of causation.

9.3 Average Glucose and Stroke

```
df.groupby("stroke")["avg_glucose_level"].agg(
    ["count", "mean", "median", "std", "min", "max"]
)
```

Stroke	Count	Mean Glucose	Median	Std Dev	Min	Max
0	4861	104.80	91.47	43.85	55.12	267.76
1	249	132.54	105.22	61.92	56.11	271.74

Observed metric: mean glucose is approximately 132.54 for stroke cases compared with 104.80 for non-stroke cases.

9.4 BMI and Stroke

```
df.groupby("stroke")["bmi"].agg(
    ["count", "mean", "median", "std", "min", "max"]
)
```

Stroke	Count	Mean BMI	Median	Std Dev	Min	Max
0	4861	28.83	28.30	7.78	10.30	97.60
1	249	30.22	28.89	5.83	16.90	56.60

Observed metric: the stroke group has a mean BMI of approximately 30.22 compared with 28.83 for non-stroke records. The difference is smaller than the observed age and glucose differences.

10. Categorical Bivariate Analysis

```
cross_smoking = pd.crosstab(
    df["smoking_status"], df["stroke"], margins=True
)
display(cross_smoking)

cross_hyper = pd.crosstab(
    df["hypertension"], df["stroke"], normalize="index"
) * 100
display(cross_hyper.round(2))
```

10.1 Hypertension vs Stroke

Hypertension	No Stroke	Stroke	Stroke Rate
0	4429	183	3.97%
1	432	66	13.25%

The observed stroke rate is approximately 3.97% among patients without hypertension and 13.25% among patients with hypertension.

10.2 Heart Disease vs Stroke

Heart Disease	No Stroke	Stroke	Stroke Rate
0	4632	202	4.18%
1	229	47	17.03%

The observed stroke rate is approximately 4.18% among patients without heart disease and 17.03% among patients with heart disease.

10.3 Smoking Status vs Stroke

Smoking Status	Patients	Stroke Cases	Observed Stroke Rate
Unknown	1544	47	3.04%
formerly smoked	885	70	7.91%
never smoked	1892	90	4.76%
smokes	789	42	5.32%

Among the observed categories, formerly smoked has the highest stroke rate at approximately 7.91%, followed by smokes at 5.32%, never smoked at 4.76%, and Unknown at 3.04%. These are descriptive group rates, not causal effects.

11. Multivariate Analysis

11.1 Pair Plot

```
pair_cols = [
    "age", "avg_glucose_level", "bmi",
    "hypertension", "heart_disease", "stroke"
]

sns.pairplot(
    df[pair_cols],
    hue="stroke",
    corner=True,
    plot_kws={"alpha":0.35}
)
plt.show()
```

The pair plot displays multiple pairwise relationships in one visualization. The stroke color allows visual comparison of distributions and relationships between stroke groups.

11.2 Multivariate Risk Profile

```
risk_profile = df.groupby("stroke").agg(
    patients=("id", "count"),
    avg_age=("age", "mean"),
    avg_glucose=("avg_glucose_level", "mean"),
    avg_bmi=("bmi", "mean"),
    hypertension_rate=("hypertension", "mean"),
    heart_disease_rate=("heart_disease", "mean")
)

risk_profile["hypertension_rate"] *= 100
risk_profile["heart_disease_rate"] *= 100
```

Stroke	Patients	Avg Age	Avg Glucose	Avg BMI	Hypertension Rate	Heart Disease Rate
0	4861	41.97	104.80	28.83	8.89%	4.71%
1	249	67.73	132.54	30.22	26.51%	18.88%

12. Group-Based Analysis

12.1 Stroke Rate by Age Group

```
df["age_group"] = pd.cut(
    df["age"],
    bins=[-np.inf, 18, 40, 60, 75, np.inf],
    labels=["<=18", "19-40", "41-60", "61-75", "76+"]
)

age_group_summary = df.groupby(
    "age_group", observed=False
).agg(
    patients=("id", "count"),
    strokes=("stroke", "sum"),
    stroke_rate=("stroke", "mean")
)

age_group_summary["stroke_rate"] *= 100
```

Age Group	Patients	Stroke Cases	Stroke Rate
<=18	916	2	0.22%
19-40	1328	6	0.45%
41-60	1562	64	4.10%
61-75	839	81	9.65%
76+	465	96	20.65%

The observed stroke rate rises across the age groups in this dataset: approximately 0.22% for <=18, 0.45% for 19–40, 4.10% for 41–60, 9.65% for 61–75, and 20.65% for 76+.

12.2 Stroke Rate by Work Type

Work Type	Patients	Stroke Cases	Stroke Rate
Govt_job	657	33	5.02%
Never_worked	22	0	0.00%
Private	2925	149	5.09%
Self-employed	819	65	7.94%
children	687	2	0.29%

Self-employed has the highest observed stroke rate among the listed work categories at 7.94%, followed by Private at 5.09% and Govt_job at 5.02%. Never_worked has 0 cases in this dataset, while children has 0.29%.

13. Pivot Table Analysis

13.1 Stroke by Gender

```
pivot_gender = pd.pivot_table(  
    df, index="gender", columns="stroke",  
    values="id", aggfunc="count", fill_value=0  
)  
display(pivot_gender)
```

Gender	No Stroke	Stroke
Female	2853	141
Male	2007	108
Other	1	0

13.2 Stroke by Work Type

Work Type	No Stroke	Stroke
Govt_job	624	33
Never_worked	22	0
Private	2776	149
Self-employed	754	65
children	685	2

13.3 Residence Type and Hypertension

This pivot table combines residence type and hypertension. count measures observations and sum measures stroke cases because stroke is coded 0/1.

Residence	Hypertension	Patients	Stroke Cases
Urban	0	2349	101
Urban	1	247	34
Rural	0	2263	82
Rural	1	251	32

14. Crosstab Analysis

14.1 Hypertension vs Stroke

```
ct_hyper = pd.crosstab(  
    df["hypertension"], df["stroke"], margins=True  
)  
display(ct_hyper)
```

```
display(
    pd.crosstab(
        df["hypertension"], df["stroke"],
        normalize="index"
    ).mul(100).round(2)
)
```

crosstab creates a frequency table. normalize='index' changes each row into percentages, so the stroke rate can be compared within each hypertension group.

14.2 Heart Disease vs Stroke

The same method is applied to heart_disease. The observed stroke rate is 17.03% for heart-disease=1 compared with 4.18% for heart-disease=0.

14.3 Smoking Status vs Stroke

The smoking crosstab provides both counts and row percentages. This supports comparison of observed stroke rates among smoking categories.

15. Correlation & Relationship Analysis

```
corr_cols = [
    "age", "hypertension", "heart_disease",
    "avg_glucose_level", "bmi", "stroke"
]
```

```
corr = df[corr_cols].corr()
display(corr.round(3))
```

Variable vs Stroke	Correlation
Age	0.245
Hypertension	0.128
Heart disease	0.135
Average glucose	0.132
BMI	0.039

The strongest listed linear correlation with stroke is age at approximately 0.245. Hypertension is approximately 0.128, heart disease 0.135, average glucose 0.132, and BMI 0.039. These are correlation measures, not proof of causation.

```
corr_pairs = (
    corr.where(np.triu(np.ones(corr.shape), k=1).astype(bool))
    .stack()
    .sort_values(key=lambda s: s.abs(), ascending=False)
)
```

```
display(corr_pairs.to_frame("correlation").head(10))
```

This code keeps only one copy of each pair, removes the lower-triangle duplicates, stacks the matrix into a Series, and sorts relationships by absolute correlation strength.

16. Additional Visual Storytelling – Business Questions

Question 1: How does stroke rate vary by age group?

The age-group analysis shows a clear increasing pattern in observed stroke rate as age group increases. The 76+ group has the highest observed rate at 20.65%.

Question 2: Which smoking categories have the highest observed stroke rate?

Formerly smoked has the highest observed stroke rate at 7.91%, followed by smokes at 5.32%. This is descriptive and should not be interpreted as a causal smoking effect.

Question 3: How do hypertension and heart disease jointly relate to stroke?

Hypertension	Heart Disease	Observed No-Stroke %	Observed Stroke %
0	0	96.61%	3.39%
0	1	83.96%	16.04%
1	0	87.79%	12.21%
1	1	79.69%	20.31%

The highest observed stroke rate among these four combinations is the group with both hypertension=1 and heart_disease=1, at approximately 20.31%. Again, this is an observed association within the dataset.

17. Key Business Insights

- The cleaned dataset contains 5,110 patients and 12 variables with zero missing values and zero duplicate rows.
- Stroke is highly imbalanced: 249 cases (4.87%) versus 4,861 non-stroke cases (95.13%).
- The stroke group has a higher mean age (67.73) than the non-stroke group (41.97).
- The stroke group has a higher mean glucose level (132.54) than the non-stroke group (104.80).
- The observed stroke rate rises from 0.22% in the <=18 group to 20.65% in the 76+ group.
- Observed stroke rate is 13.25% for hypertension=1 versus 3.97% for hypertension=0.
- Observed stroke rate is 17.03% for heart_disease=1 versus 4.18% for heart_disease=0.
- Age has the strongest correlation with stroke among the selected numerical variables (approximately 0.245).

18. Recommendations

- Use age, glucose, hypertension, heart disease, BMI and smoking status as candidate variables for further investigation.
- Prioritize analysis of patient groups with higher observed stroke rates, subject to clinical validation.
- Use class-imbalance-aware evaluation in future predictive modeling.
- Combine multiple risk factors rather than relying on a single variable.
- Treat EDA findings as associations that guide further analysis, not as medical diagnoses or causal conclusions.
- Validate operational healthcare recommendations with qualified clinical professionals and appropriate statistical testing.

19. Sprint 2 Metric Dashboard

Metric	Value
Total patients	5,110
Variables	12
Missing values	0
Duplicate rows	0
Stroke cases	249
Non-stroke cases	4,861
Overall stroke rate	4.87%
Mean age – stroke	67.73
Mean age – no stroke	41.97
Mean glucose – stroke	132.54
Mean glucose – no stroke	104.80
Mean BMI – stroke	30.22
Mean BMI – no stroke	28.83
Highest age-group stroke rate	20.65% (76+)
Highest smoking-category rate	7.91% (formerly smoked)
Hypertension=1 stroke rate	13.25%
Heart disease=1 stroke rate	17.03%
Highest joint rate	20.31% (hypertension=1, heart disease=1)
Strongest selected correlation with stroke	Age: 0.245

20. Sprint 2 Questions and Answers

Q1. What is the goal of Sprint 2?

Answer: Discover meaningful patterns, trends and relationships in the cleaned stroke dataset through EDA and visualization.

Q2. How many records are in the cleaned dataset?

Answer: 5,110.

Q3. How many variables are present?

Answer: 12.

Q4. Are missing values present?

Answer: No. The cleaned dataset has 0 missing cells.

Q5. Are duplicate rows present?

Answer: No. Duplicate count is 0.

Q6. How many stroke cases are present?

Answer: 249 (4.87%).

Q7. Is the target balanced?

Answer: No. The stroke class is strongly imbalanced.

Q8. What is the mean age of stroke patients?

Answer: 67.73 years.

Q9. What is the mean glucose level of stroke patients?

Answer: 132.54.

Q10. What is the observed stroke rate for hypertension=1?

Answer: 13.25%.

Q11. What is the observed stroke rate for heart disease=1?

Answer: 17.03%.

Q12. Which age group has the highest observed stroke rate?

Answer: 76+ with 20.65%.

Q13. Which smoking category has the highest observed stroke rate?

Answer: Formerly smoked with 7.91%.

Q14. Which selected variable has the strongest correlation with stroke?

Answer: Age, with correlation ≈ 0.245 .

Q15. What does a correlation of 0.245 mean?

Answer: It indicates a positive linear association of modest strength in this dataset; it does not establish causation.

Q16. Why use crosstab?

Answer: To compare category combinations and calculate within-group counts or percentages.

Q17. Why use pivot tables?

Answer: To summarize multiple dimensions using grouped counts, sums, means or medians.

Q18. Why use box plots?

Answer: To compare distributions, medians, spread and potential extreme observations across groups.

Q19. Why is class imbalance important?

Answer: A minority stroke class can make accuracy misleading, so future models need appropriate evaluation metrics.

Q20. What is the next stage?

Answer: Statistical testing and/or predictive modeling for stroke-risk classification, as stated in the Sprint 2 notebook.

21. Sprint 2 Conclusion

Sprint 2 applies descriptive statistics, univariate analysis, bivariate analysis, multivariate analysis, group-based analysis, pivot tables, crosstabs, correlation analysis, visualization and business storytelling to the cleaned healthcare stroke dataset.

The analysis shows 5,110 patients, 249 stroke cases, and a 4.87% observed stroke rate. Age, average glucose level, hypertension and heart disease show notable observed differences between stroke and non-stroke groups. The age-group analysis also shows increasing observed stroke rates across older groups.

These findings provide a foundation for the next stage. The notebook specifically states that the next stage can involve statistical testing and/or predictive modeling for stroke-risk classification. EDA findings should continue to be treated as associations rather than causal or clinical conclusions.

22. Code-to-Metric Traceability

Sprint 2 Topic	Main Code/Function	Metric Produced
Dataset loading	pd.read_csv()	Rows, columns
Data quality	isnull().sum(), duplicated().sum()	Missing cells, duplicates
Descriptive statistics	describe(), median(), var(), skew()	Mean, median, spread, skewness
Univariate numerical	histplot(), boxplot()	Distribution/outlier view
Categorical analysis	value_counts()	Counts and percentages
Target analysis	value_counts(normalize=True)	Stroke rate
Bivariate numerical	corr(), groupby()	Correlation and group differences
Multivariate	pairplot(), groupby().agg()	Combined risk profile
Group analysis	pd.cut(), groupby()	Age-group and category stroke rates
Pivot tables	pd.pivot_table()	Multi-dimensional summaries
Crosstabs	pd.crosstab()	Counts and row percentages
Correlation	DataFrame.corr()	Relationship strength
Business storytelling	bar plots + summaries	Decision-oriented insights